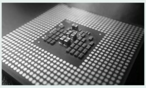
Kubeflow takes ML to Kubernetes

The newly announced project from Google engineers, called Kubeflow, aims to leverage machine learning to address the hurdles of launching convoluted workloads on Kubernetes, which is an open source platform that serves as the backbone of container orchestration management. With the arrival of Kubeflow, Kubernetes will be able to use machine learning (ML) stacks anywhere.

Specifically, Kubeflow includes the JupyterHub platform, which enables data science and research groups to create and manage Jupyter notebook servers. Additionally, Kubeflow includes a TensorFlow Customer Resource, which can support either CPUs or GPUs, and be tailored to manage a specific container cluster size.



In a company blog post, Philip Winder, an engineer and consultant at Container Solutions, wrote, "Like DevOps has merged operations and development, DataDevOps will consume data science."

The company also shared that it believes working with multiple environments, from development to production, will become the norm for most Kubeflow users. Consequently, Kubernetes is making use of the Ksonnet project, which is intended to make it easier to transfer workloads across multiple environments.

Kubernetes is currently working to cultivate a community around the project. Among the companies collaborating on the project are CaiCloud, Red Hat and OpenShift, Canonical, Weaveworks, Container Solutions, etc. The project was much needed to make it easier to set up and productionise machine learning workloads on Kubernetes.

To mark the anniversary, OSI is organising several activities through the year. Opensource.net was launched for this purpose. Nick Vidal from the OSI informed that the successful completion of two decades will witness worldwide celebrations in conjunction with major technology conferences such as FOSDEM, OSCON, Open Source Summit, FOSSAsia, Campus Party, Linux.conf.au, etc.

Baidu's Apollo Autonomous driving platform chooses ON Semiconductor's image sensors

Image sensing is a key component of the Apollo platform, which supports enhanced autonomous driving. Post collaboration with ON Semiconductor, the partners of Baidu's Apollo Autonomous driving platform can get the jointly developed plug-and-play compatible imaging solutions.

Apollo provides an open source, reliable software and hardware system, enabling the efficient development of autonomous driving systems by automotive systems designers. The collaboration will simplify the implementation of image sensing solutions for automotive manufacturers and suppliers, while enhancing the speed.

ON Semiconductor's fully-qualified 3mm-based advanced CMOS image sensors provide flexibility to move to future sensors at volume deployment. It also allows customers to begin development of vision systems for autonomous driving. With high dynamic range (HDR), the sensor is able to provide crisp, clear single images as well as video in the challenging low and mixed light scenes typical of automotive environments.

In addition to the simplified and accelerated design, and test and implementation of automotive camera applications, the relationship will offer all Apollo partners the early opportunities to work with future generation, breakthrough image sensing technologies from ON Semiconductor.

Ross Jatou, VP and GM of the automotive solutions group at ON Semiconductor said, "We believe that the value of such a platform to automotive system designers will be tremendous. Image sensors are fundamental components of ADAS implementations throughout the vehicle, and they will become even more relevant as the industry moves towards fully autonomous cars. Joining forces with Baidu by providing the image sensor solution for the Apollo platform is further validation of ON Semiconductor's leading position in automotive image sensing."

'Indian Open Source Project Incubator' by Paytm

Paytm has launched a virtual incubator to promote its 'Build for India' initiative. Now professionals and students can build open source based solutions, and share these projects with the global developer community, thanks to the recent announcement of the 'Indian Open Source Project Incubator' from Paytm.

According to Vijay Shekhar Sharma, founder and CEO, Paytm, the virtual incubator will aim to find developers who can use technology and design to build world class solutions. The company shared that as part of the incubator programme, Paytm will make its premises available for mentoring, meet-ups, as well as technology and design support, from time to time.

The launch is in line with the company's 'Build for India' initiative to promote a 'building and sharing' culture among developers in the country. It will focus on open source projects in the field of education, digital lifestyle, and financial